

# Exponentials & Logarithms



## Evaluating exponential expressions

- 1 Evaluate  $2^5$ .
  - 2 Evaluate  $3^{-2}$ .
  - 3 Evaluate  $16^{1/2}$ .
  - 4 Evaluate  $27^{2/3}$ .
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## Evaluating logarithms

- 5 Evaluate  $\log_2 32$ .
  - 6 Evaluate  $\log_3 81$ .
  - 7 Evaluate  $\log_5 1$ .
  - 8 Evaluate  $\log_{10} 0.01$ .
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## Log laws: product and quotient

- 9 Write  $\log_2 (8x)$  as a sum of logarithms.
  - 10 Write  $\log_3 \left(\frac{x}{9}\right)$  as a difference of logarithms.
  - 11 Combine  $\log_5 5 + \log_5 25$  into a single logarithm and evaluate.
  - 12 Combine  $\log_2 24 - \log_2 3$  into a single logarithm and evaluate.
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## Log laws: the power rule

- 13 Write  $\log(x^5)$  using the power rule.
- 14 Evaluate  $\log_2 (4^3)$  using the power rule.
- 15 Write  $2\log_3 x + \log_3 y$  as a single logarithm.

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### Expanding and condensing logarithmic expressions

- 16 Expand  $\log\left(\frac{x^3y}{z^2}\right)$  fully.
- 17 Condense  $3\ln x - 2\ln y$  into a single logarithm.
- 18 Expand  $\log_2(\sqrt{x} \cdot y^3)$  fully.

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### Solving exponential equations (matching bases)

- 19 Solve  $2^x = 32$ .
- 20 Solve  $3^{x+1} = 81$ .
- 21 Solve  $5^{2x} = 125$ .

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### Solving exponential equations using logarithms

- 22 Solve  $2^x = 10$  for  $x$ , rounded to three decimal places.
- 23 Solve  $5^x = 40$  for  $x$ , rounded to three decimal places.
- 24 Solve  $3^{2x} = 50$  for  $x$ , rounded to three decimal places.
- 25 Solve  $e^x = 15$  for  $x$ , rounded to three decimal places.

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### Solving logarithmic equations

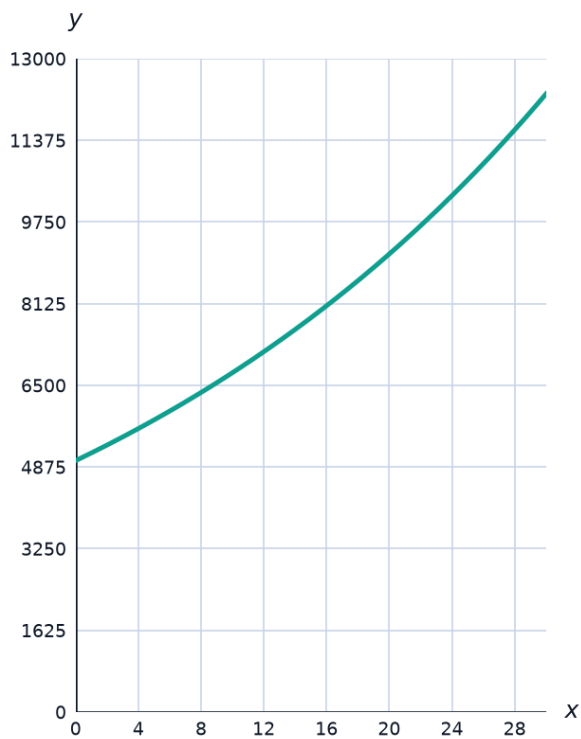
- 26 Solve  $\log_2(x) = 5$ .
- 27 Solve  $\log_3(x - 1) = 2$ .
- 28 Solve  $\log(x) + \log(x - 3) = 1$ .
- 29 Solve  $\log_2(x + 3) + \log_2(x - 1) = 5$ .
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## Natural logarithms and e

- 30 Evaluate  $\ln(e^3)$ .
- 31 Evaluate  $\ln(1)$ .
- 32 Solve  $\ln(x) = 2$  for  $x$ , in terms of  $e$ .
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## Exponential growth and decay models

- 33 The population of a town is modeled by  $P(t) = 5000e^{0.03t}$ , where  $t$  is in years. Find  $P(10)$ , rounded to the nearest whole number.



- 34 A radioactive substance decays according to  $A(t) = 200e^{-0.05t}$  (grams). Find  $A(20)$ , rounded to one decimal place.
- 35 Determine whether  $N(t) = 50e^{0.02t}$  models growth or decay, and justify your answer.
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## Word problems: exponential models in context

- 36** This question has two parts. An investment of 2000 dollars grows according to  $A(t) = 2000e^{0.04t}$ , where  $t$  is in years.
- a** Find the value of the investment after 5 years, rounded to the nearest dollar.
  - b** Find how many years it takes for the investment to double, rounded to one decimal place (using  $\ln 2 \approx 0.693$ ).
- 37** This question has two parts. A cup of coffee cools according to  $T(t) = 20 + 60e^{-0.1t}$  ( $^{\circ}\text{C}$ ), where  $t$  is minutes after brewing and  $20^{\circ}\text{C}$  is room temperature.
- a** Find the temperature after 15 minutes, rounded to one decimal place (using  $e^{-1.5} \approx 0.2231$ ).
  - b** Find how many minutes it takes for the temperature to drop to  $35^{\circ}\text{C}$ , rounded to one decimal place (using  $\ln 4 \approx 1.386$ ).